

Momentum Conservation in Collisions

PhET Collision Lab · PhET

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/collision-lab>
Curriculum links: momentum · forces · motion

Learning intentions

- I can calculate the momentum of an object before and after a collision.
- I can predict the outcome of elastic and inelastic collisions using conservation of momentum.
- I can distinguish between elastic and inelastic collisions using kinetic energy data.

Getting started

Open PhET Collision Lab and set up a simple 1D collision between two balls. Use the simulation's controls to set each ball's mass and initial velocity, and turn on the values and momenta displays if available so you can read off numbers before and after the collision.

Tasks

1. Set up a collision where a 1 kg ball moving at 2 m/s hits a stationary 1 kg ball. Predict what will happen before you run it.

2. Calculate the total momentum of the system before the collision from the previous task (momentum = mass × velocity). Run the collision and record the velocities afterward — does total momentum still match your calculation?

Working:

3. Run three trials with different masses and starting velocities, and record the velocities before and after each collision.

Trial	Mass 1 & 2 (kg)	Velocity before (m/s)	Velocity after (m/s)

4. Set the elasticity to 100% (perfectly elastic) for one trial and 0% (perfectly inelastic) for another, keeping masses and starting velocities the same. Describe the difference in what happens after the collision.

5. For your perfectly inelastic trial, calculate the kinetic energy of the system before and after the collision. Is kinetic energy conserved?

Working:

6. For your perfectly elastic trial, calculate the kinetic energy before and after. Compare this to your inelastic result.

Working:

7. Explain why momentum is conserved in both elastic and inelastic collisions, but kinetic energy is only conserved in elastic collisions.

8. A loaded truck and a small car collide head-on at the same speed. Using conservation of momentum, explain qualitatively why the smaller car experiences a much larger change in velocity than the truck.

Extension (optional)

Set up a 2D collision (balls colliding at an angle rather than head-on). Explain what stays the same and what becomes more complex when you move from analysing 1D collisions to 2D collisions.
